

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks:25 Annexure A

Case Study

Code of Conduct:

I GAYATHRI N P (192421362) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

ANSWER:

Introduction:

A hospital can develop an intelligent disease diagnosis system using Bayes' Theorem, Naive Bayes Classifier, and Bayesian Belief Networks. The system predicts the probability of a disease based on patient symptoms, medical history, and test results.

Bayes' Theorem:

Bayes' Theorem is used to calculate the probability of a disease when some symptoms or evidence are observed.

$$P(D|S) = P(S|D) \times P(D) / P(S)$$

Where:

$P(D|S)$ = Probability of disease given the symptoms

$P(S|D)$ = Probability of symptoms given the disease

$P(D)$ = Prior probability of the disease

$P(S)$ = Probability of observing the symptoms

For example, if a patient has fever, cough, and breathing difficulty, the system calculates the probability of diseases such as pneumonia or flu.

Naive Bayes Classifier:

Naive Bayes assumes that the symptoms are conditionally independent when the disease is known.

$$P(D|S_1, S_2, \dots, S_n) \text{ is proportional to } P(D) \times P(S_1|D) \times P(S_2|D) \times \dots \times P(S_n|D)$$

The steps are:

1. Collect patient symptoms and medical history.
2. Calculate the prior probability of each disease.
3. Calculate the probability of each symptom for every disease.
4. Calculate the posterior probability.
5. Select the disease with the highest probability.

Bayesian Belief Network:

A Bayesian Belief Network represents relationships between diseases, symptoms, and risk factors using a directed graph.

Example:

Smoking -> Lung Disease -> Cough -> Breathing Problem

A Bayesian Belief Network can represent dependencies between variables. For example, smoking may increase the probability of lung disease, which may cause coughing and breathing problems.

Handling Uncertainty:

Medical diagnosis involves uncertainty because different diseases can have similar symptoms. Patient information may also be incomplete or inaccurate.

Probability helps the system represent uncertainty.

Example:

Pneumonia = 0.70

Flu = 0.20

Other diseases = 0.10

Therefore, the system predicts pneumonia because it has the highest probability.

Role of Probability Learning:

The probabilities can be learned from historical medical data. For example:

$P(\text{Fever}|\text{Pneumonia}) = \text{Number of pneumonia patients with fever} / \text{Total number of pneumonia patients}$

As more patient data becomes available, the probability estimates can be updated.

Real-World Constraints:

1. Missing patient information
2. Noisy medical data
3. Limited training data
4. Privacy of patient information
5. Bias in medical datasets
6. Different symptoms among patients
7. Computational requirements
8. Need for doctor verification

Conclusion:

A probabilistic diagnosis system can effectively handle uncertainty and incomplete medical information. Naive Bayes is simple and fast, while Bayesian Belief Networks are useful for representing relationships between symptoms and medical factors. The system should assist doctors rather than completely replace medical judgment.

2. **An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.**

Introduction:

An organization can develop a spam detection system using Naive Bayes and Bayes Optimal Classifier. The system classifies emails into two categories: Spam and Not Spam.

Naive Bayes Model:

Naive Bayes calculates the probability that an email belongs to a particular class based on the words present in the email.

$P(C|W_1, W_2, \dots, W_n)$ is proportional to $P(C) \times P(W_1|C) \times P(W_2|C) \times \dots \times P(W_n|C)$

Where C represents Spam or Not Spam.

Steps:

1. Collect labelled emails.
2. Preprocess the emails.
3. Remove unnecessary symbols and words.
4. Extract important features such as words and phrases.
5. Calculate word probabilities.
6. Train the Naive Bayes classifier.
7. Calculate Spam and Not Spam probabilities.
8. Assign the email to the class with the highest probability.

Example:

Email: "Congratulations! You won a free prize."

Words such as "free", "prize", "won", and "congratulations" may have high probabilities in spam emails. Therefore, the classifier may classify the email as Spam.

Bayes Optimal Classifier:

The Bayes Optimal Classifier considers multiple possible hypotheses and combines their predictions according to their probabilities.

$P(C|X) = \text{Sum of } P(C|X, h) \times P(h|X)$

It can provide better theoretical performance but requires more computation than Naive Bayes.

Maximum Likelihood Estimation:

Maximum Likelihood Estimation estimates model parameters by maximizing the likelihood of the training data.

$\text{Theta_MLE} = \text{argmax } P(D|\text{Theta})$

For example, if the word "free" occurs frequently in spam emails, MLE estimates a high

value for $P(\text{free}|\text{Spam})$.

Advantages of MLE:

1. Simple
2. Fast
3. Easy to implement
4. Works well with sufficient training data

Disadvantages of MLE:

1. Can overfit with small datasets.
2. Probability estimates may be unreliable when data is limited.

Minimum Description Length:

The Minimum Description Length principle selects a model that minimizes the total description length.

$$\text{MDL} = L(\text{Model}) + L(\text{Data}|\text{Model})$$

MDL balances model complexity and data fitting. It helps reduce overfitting by preferring simpler models when two models provide similar performance.

MLE and MDL Comparison:

MLE focuses mainly on maximizing the likelihood of the training data.

MDL focuses on balancing model complexity and data fitting.

MLE is simpler to implement.

MDL is useful for feature selection and controlling overfitting.

Effect of Sample Size:

With a small sample size, probability estimates may be unreliable and the model may overfit.

With a large sample size, probability estimates become more reliable and classification accuracy usually improves.

Effect of Hypothesis Space:

A large hypothesis space provides more possible models but increases computational cost and the risk of overfitting.

A smaller hypothesis space reduces complexity but may cause underfitting if it is too restrictive.

Conclusion:

Naive Bayes is a simple and efficient method for spam detection. The Bayes Optimal Classifier provides a theoretically strong approach by considering multiple hypotheses. MLE provides simple probability estimation, while MDL balances accuracy and model complexity. A suitable sample size and hypothesis space are important for achieving good classification accuracy.

3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.

Introduction:

A retail company can use the Expectation-Maximization algorithm to identify hidden customer groups from purchasing data. EM is useful when cluster labels are unknown and some information may be incomplete.

Examples of customer groups are:

1. High-value customers
2. Regular customers
3. Occasional customers

EM Algorithm:

The EM algorithm has two main steps:

1. Expectation Step
2. Maximization Step

Expectation Step:

In the Expectation step, the algorithm calculates the probability that each customer belongs to each cluster.

$$P(Z=k|X, \Theta)$$

For example:

Customer A:

Cluster 1 = 0.80

Cluster 2 = 0.15

Cluster 3 = 0.05

Customer A has the highest probability of belonging to Cluster 1.

Maximization Step:

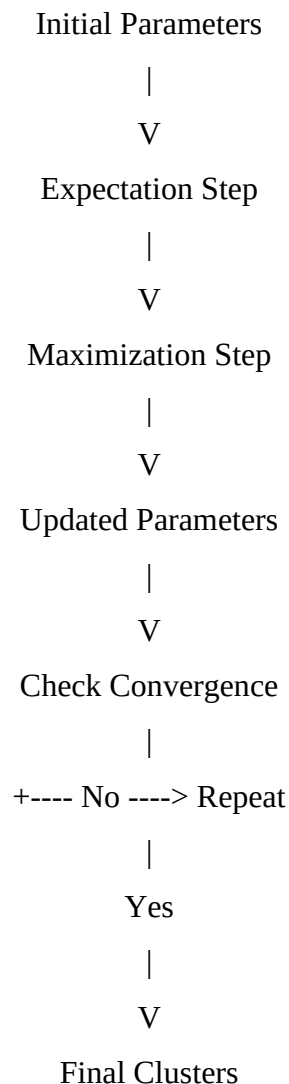
In the Maximization step, the algorithm updates the model parameters using the probabilities obtained in the Expectation step.

The parameters include:

1. Cluster means
2. Variances
3. Mixing probabilities

The E-step and M-step are repeated until the algorithm converges.

Working of EM:



Example:

Consider the following customer data:

Customer A: Spending = 9000, Visits = 15

Customer B: Spending = 8500, Visits = 13

Customer C: Spending = 2000, Visits = 4

Customer D: Spending = 2500, Visits = 5

EM may identify:

Cluster 1 = High-value customers

Cluster 2 = Low-value customers

Convergence:

EM improves the likelihood during each iteration. The algorithm stops when the change in likelihood becomes very small.

$$|L_{\text{new}} - L_{\text{old}}| < \text{Epsilon}$$

Initialization Challenges:

The initial parameters can affect the final result. Poor initialization can result in slow convergence or poor clusters.

Solutions include:

1. Random initialization
2. Multiple random restarts
3. Using K-Means for initialization

Role of Probability Learning:

EM learns the probability distribution of different customer groups. It provides a probability for each customer instead of assigning the customer strictly to one group.

Marketing Applications:

High-value customers can be given premium products and loyalty rewards.

Price-sensitive customers can be given discounts and coupons.

Inactive customers can receive special offers and re-engagement campaigns.

Limitations:

1. Sensitive to initialization
2. May converge to a local optimum
3. Requires assumptions about probability distributions
4. Choosing the number of clusters can be difficult
5. Computational cost can increase for large datasets
6. Noisy or incomplete data can affect the results

Conclusion:

The EM algorithm is useful for identifying hidden customer patterns and handling incomplete information. It provides probabilistic cluster assignments and can help companies perform targeted marketing and personalized promotions.

4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.

Introduction:

A medical image classification system can classify images into Normal and Abnormal categories using concept learning, hypothesis space search, and the Candidate Elimination algorithm.

Hypothesis Space:

The hypothesis space contains all possible rules that can be used to classify the images.

Possible features include:

1. Color
2. Texture
3. Shape
4. Size
5. Edge pattern
6. Intensity

Example hypothesis:

Texture = Irregular AND Shape = Abnormal $\rightarrow$ Abnormal

Candidate Elimination Algorithm:

Candidate Elimination maintains a set of hypotheses that are consistent with the training examples. This set is called the Version Space.

It uses two boundaries:

1. Specific Boundary (S)
2. General Boundary (G)

The Specific Boundary contains the most specific hypotheses.

The General Boundary contains the most general hypotheses.

Working:

For a positive training example, the Specific Boundary is generalized when necessary.

For a negative training example, the General Boundary is specialized to remove incorrect hypotheses.

The process continues until the version space contains hypotheses that are consistent with all training examples.

Example:

Texture = Smooth, Shape = Normal, Class = Normal

Texture = Irregular, Shape = Abnormal, Class = Abnormal

The algorithm updates the S and G boundaries based on these examples.

Finite Hypothesis Space:

A finite hypothesis space contains a limited number of possible hypotheses.

Advantages:

1. Easier search
2. Lower computational cost
3. Faster learning

Disadvantage:

The correct concept may not exist in the hypothesis space.

Infinite Hypothesis Space:

An infinite hypothesis space contains a very large or unlimited number of possible hypotheses.

Advantages:

1. Greater flexibility
2. Can represent complex concepts

Disadvantages:

1. Higher computational cost
2. Greater risk of overfitting
3. More training examples may be required

Sample Complexity:

Sample complexity is the number of training examples required to learn a concept with acceptable accuracy.

A larger hypothesis space generally requires more training examples.

A simplified relationship is:

m is proportional to $\log(|H|) / \epsilon$

Where:

m = Number of training examples

$|H|$ = Size of hypothesis space

ϵ = Acceptable error

Mistake Bound Model:

The mistake-bound model measures the number of mistakes an algorithm can make before learning the correct concept.

A smaller mistake bound indicates faster learning.

Generalization:

The model should correctly classify unseen medical images. Good generalization requires sufficient training data, suitable features, representative examples, and proper handling of noise.

Limitations:

1. Large hypothesis spaces are computationally expensive.
2. Medical images contain noise.

3. Real image features can be complex.
4. Incorrect labels can affect learning.
5. The true concept may not be present in the hypothesis space.

Conclusion:

Candidate Elimination searches the hypothesis space and maintains specific and general boundaries. Its performance depends on the quality of training data, selected features, and the hypothesis space. Although modern medical image classification commonly uses deep learning, Candidate Elimination is useful for understanding concept learning.

5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

Introduction:

A financial institution can use Bayesian inference, Gibbs sampling, and probability learning to make investment decisions under uncertain market conditions. The system updates its predictions whenever new financial data becomes available.

Bayesian Inference:

Bayesian inference updates the probability of a hypothesis when new evidence is received.

$$P(H|D) = P(D|H) \times P(H) / P(D)$$

Where:

$P(H)$ = Prior probability

$P(D|H)$ = Likelihood

$P(H|D)$ = Posterior probability

$P(D)$ = Probability of data

For example, suppose the initial probability that a stock price will increase is 0.60. After receiving positive market information, the posterior probability may increase to 0.80.

Gibbs Algorithm:

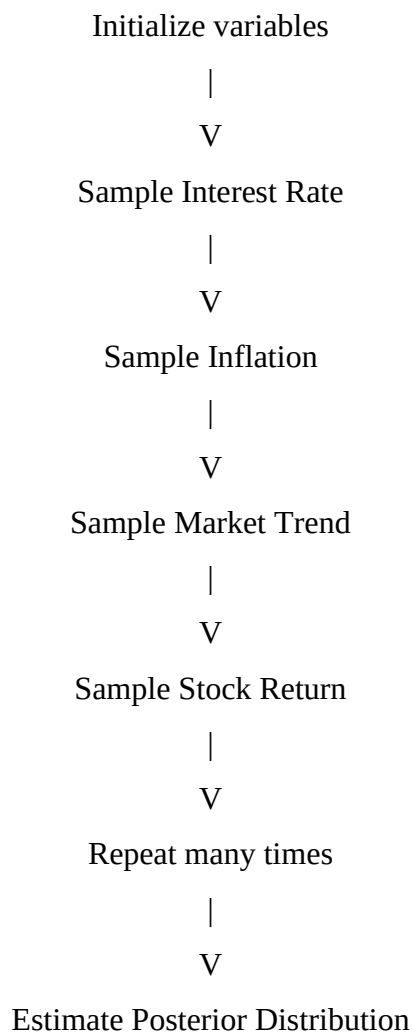
Gibbs sampling is a Markov Chain Monte Carlo method used to estimate complex probability distributions.

Suppose the model contains:

1. Interest Rate
2. Inflation
3. Market Trend
4. Stock Return

Gibbs sampling repeatedly samples each variable based on the current values of the other variables.

Working:



Role of Prior Knowledge:

Prior knowledge can include:

1. Historical stock performance
2. Interest-rate trends
3. Economic indicators
4. Company performance
5. Previous market information

Prior knowledge influences predictions, especially when the amount of new data is small.

Risk Evaluation:

Posterior probabilities can be used to estimate investment risk.

Example:

Probability of high return = 0.75

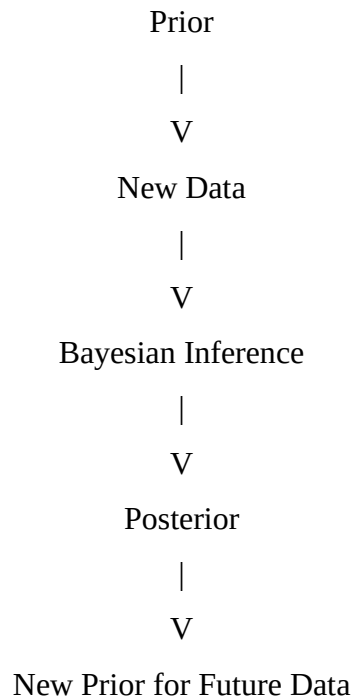
Probability of moderate return = 0.15

Probability of loss = 0.10

The financial institution can use these probabilities to compare potential returns and risks.

Dynamic Updating:

One important advantage of Bayesian inference is that the model can update its beliefs when new data arrives.



Therefore, the model can adapt to changing market conditions.

Computational Challenges:

1. High-dimensional financial data
2. Large number of Gibbs sampling iterations
3. Slow convergence
4. Correlation between variables
5. Large computational requirements

Optimization Strategies:

1. Use parallel processing.
2. Use better initialization.
3. Remove the burn-in period.
4. Monitor convergence.
5. Reduce unnecessary features.
6. Use efficient sampling techniques.

Evaluation:

The system can be evaluated using:

1. Prediction accuracy
2. Mean Absolute Error
3. Mean Squared Error
4. Expected return
5. Risk-adjusted return
6. Posterior prediction accuracy

Conclusion:

Bayesian inference provides a powerful method for making investment decisions under uncertainty. Gibbs sampling helps estimate complex posterior distributions, while probability learning allows the model to update its predictions using new data. However, market volatility and computational complexity must be considered when applying the system in real-world financial applications.